

Nathaniel J. Park

PRINCIPAL CLOUD ARCHITECT · DEVSECOPS ·
PLATFORM ENGINEERING

Email:

nathaniel.park@cloudarch.dev

Phone: +1 (617) 403-9928

Location: Boston, MA 02101

LinkedIn:

linkedin.com/in/nathaniel-park-arch

GitHub: github.com/njpark-cloud

Current Role: Principal Cloud Architect @
Fidelity Investments

Experience:
16+ Years

Clearance: None
(Available)

Relocation:
Open

Work Auth:
US Citizen

16+	\$85M	350+	15	99.99%	4000+
YEARS EXPERIENCE	INFRA COST SAVED	CLOUD MIGRATIONS	CERTIFICATIONS	SLA ACHIEVED	ENGINEERS ENABLED

PROFESSIONAL SUMMARY

Principal Cloud Architect and DevSecOps leader with **16+ years** designing, implementing, and governing enterprise-scale cloud infrastructure across AWS, Azure, and GCP. Specializes in multi-cloud strategy, Kubernetes at scale, FinOps optimization, and building platform engineering cultures that enable thousands of developers. Led cloud transformations at Fidelity Investments, Akamai Technologies, HubSpot, and Rackspace. Author of 3 AWS whitepapers, 2 O'Reilly courses, and speaker at KubeCon, AWS re:Invent (2021, 2022, 2023), and HashiConf. Deep expertise in zero-trust security, hybrid cloud architecture, disaster recovery, and developer experience (DevEx). Known for translating complex technical requirements into business outcomes and mentoring next-generation infrastructure engineers.

CONTACT

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- 📍 47 Beacon St, Boston, MA 02101
- ▶ linkedin.com/in/nathaniel-park-arch
- 🔗 github.com/njpark-cloud
- 🌐 njpark.io (Portfolio)
- 💎 @njpark_cloud (Twitter/X)

TECHNICAL SKILLS

CLOUD PLATFORMS

AWS	■	98%
Azure	■	92%
GCP	■	85%

CONTAINER & ORCHESTRATION

Kubernetes	●	97%
Docker/Containerd	■	95%
Helm / Kustomize	■	93%

IAC & AUTOMATION

Terraform	●	98%
Pulumi	■	88%
Ansible	■	90%

CI/CD PIPELINES

GitHub Actions	●	96%
ArgoCD / Flux	●	94%
Jenkins / Tekton	■	88%

WORK EXPERIENCE

Principal Cloud Architect Fidelity Investments

Jan 2021 - Present Boston, MA

- ▶ Design and govern Fidelity's multi-cloud architecture spanning AWS (primary), Azure (DR/analytics), and on-premise data centers supporting 380 critical financial applications with 99.99% SLA; manage \$52M annual cloud budget
- ▶ Led \$42M cloud modernization initiative migrating 120 legacy Java/mainframe applications to cloud-native microservices; delivered on time and 8% under budget, achieving 40% OPEX reduction (\$14M/year savings)
- ▶ Built enterprise Kubernetes platform (EKS) on AWS supporting 4,000+ daily deployments across 200 development teams; reduced average deployment time from 45 minutes to under 4 minutes
- ▶ Architected FinOps practice reducing cloud spend from \$18M/month to \$10.8M/month via rightsizing, spot instances, Savings Plans, and automated idle resource reclamation (\$85M 5-year savings projection)
- ▶ Designed zero-trust security architecture using Istio service mesh, HashiCorp Vault, and OPA policies, meeting FINRA, SOC 2 Type II, PCI-DSS Level 1, and ISO 27001 compliance requirements
- ▶ Led development of internal IDP (Internal Developer Platform) built on Backstage, adopted by 2,200+ engineers; reduced onboarding time from 3 weeks to 2 days and eliminated 78% of infrastructure tickets
- ▶ Established Site Reliability Engineering practice: defined SLOs/SLIs for 40 critical services, implemented error budget policies, and achieved 35% reduction in P0/P1 incidents YoY
- ▶ Mentored 8 senior engineers to staff/principal level promotions; co-led internal Cloud Architecture Guild of 120 members across engineering org

AWS EKS Terraform Istio ArgoCD Vault Backstage Python Go Prometheus
Datadog

Senior Platform Engineer → Staff Architect Akamai Technologies

Mar 2017 - Dec 2020 Cambridge, MA

- ▶ Designed Akamai's internal CI/CD platform handling 50,000+ daily deployments across 350,000+ edge servers in 130 countries; reduced release cycle from weekly to multiple times per day
- ▶ Rebuilt service mesh layer using Envoy Proxy + Istio; reduced P99 inter-service latency by 41% and eliminated 99% of manual TLS certificate rotations through automated cert-manager integration

PROGRAMMING
LANGUAGES

Python		90%
Go (Golang)		82%
Bash/Shell		95%

OBSERVABILITY

Prometheus

Grafana

Datadog

Jaeger

ELK Stack

OpenTelemetry

NETWORKING & SECURITY

Istio	Envoy
Vault (HashiCorp)	Cilium
Zero Trust	mTLS

DATABASES & MESSAGING

PostgreSQL	Redis	Kafka
RabbitMQ	DynamoDB	
Cassandra		

CERTIFICATIONS

★ AWS Solutions Architect - Professional Amazon Web Services · 2022
★ AWS DevOps Engineer - Professional Amazon Web Services · 2021
★ AWS Security Specialty Amazon Web Services · 2021
★ CKA - Kubernetes Admin CNCF / Linux Foundation · 2022
★ CKS - Kubernetes Security CNCF / Linux Foundation · 2023
★ Google Professional Cloud Architect Google Cloud · 2022
★ Azure Solutions Architect Expert Microsoft · 2020
★ Terraform Associate HashiCorp · 2021
★ Vault Associate HashiCorp · 2021
★ CKAD - App Developer CNCF · 2020

EDUCATION

M.S. Computer Engineering Northeastern University Boston, MA · 2010 Specialization: Distributed Systems GPA: 3.92 / 4.0
B.E. Computer Systems University of Illinois, Chicago Chicago, IL · 2008 Magna Cum Laude · Dean's List

LANGUAGES

English	Native
Korean	Fluent (C1)
Japanese	Conversational (B1)
Spanish	Basic (A2)

- ▶ Led Terraform IaC transformation across the organization; codified 400+ reusable infrastructure modules, achieving 100% infrastructure-as-code coverage for all production environments
- ▶ Built chaos engineering program (Netflix Chaos Monkey model adapted for Akamai scale); ran 300+ automated fault injection experiments identifying 12 critical resilience gaps before they became outages
- ▶ Designed and implemented multi-region active-active Cassandra cluster for 280TB telemetry data; achieved 99.999% availability with automated failover tested quarterly
- ▶ Promoted from Senior Platform Engineer to Staff Architect in 18 months; recognized for leading cross-team initiatives worth \$28M in infrastructure efficiency gains

Kubernetes Envoy Istio Terraform Cassandra Jenkins Python Chaos Engineering

Lead DevOps Engineer HubSpot, Inc. Aug 2014 - Feb 2017 Cambridge, MA

- ▶ Drove HubSpot's migration from 800+ bare-metal servers to AWS; reduced provisioning time from 2 weeks to 20 minutes; saved \$4.2M annually in hardware/colocation costs
- ▶ Built comprehensive observability platform on ELK + Prometheus + Grafana; reduced mean time to resolution (MTTR) from 4 hours to 18 minutes for P1 incidents
- ▶ Implemented GitOps workflow with ArgoCD for Kubernetes workloads; increased deployment frequency from 2x/week to 15x/day while maintaining 99.9% deployment success rate
- ▶ Designed auto-scaling architecture that handled HubSpot's 10x traffic spikes during product launches with zero downtime and 22% reduction in compute costs
- ▶ Led a team of 6 DevOps engineers; established on-call rotation, runbook standards, and post-mortem culture that became HubSpot engineering-wide best practice

AWS Docker ELK ArgoCD Prometheus Grafana Ansible

Cloud Solutions Engineer Rackspace Jun 2011 - Jul 2014 San Antonio, TX

- ▶ Designed and implemented cloud infrastructure solutions for 220+ enterprise customers spanning e-commerce, media, healthcare, and financial services verticals
- ▶ Developed automated capacity planning and cost optimization tooling reducing customer over-provisioning by 30% across managed hosting portfolio (\$18M savings for clients)
- ▶ Achieved Rackspace Certified Expert status (top 5% of engineers) within 18 months; awarded "Fanatical Support Champion" Q3 2013

OpenStack Linux Python MySQL Nginx HAProxy

Systems Administrator TeleCommunication Sep 2008 - May 2011 Annapolis, MD

- ▶ Administered Linux/Windows hybrid infrastructure supporting 911/public-safety communications platforms deployed in 200+ US municipalities
- ▶ Implemented automated backup and disaster recovery system achieving RTO of 4 hours / RPO of 15 minutes for critical 911 dispatch systems

KEY PROJECTS

Enterprise IDP @ Fidelity

2022 - 2023

Built Fidelity's Internal Developer Platform on Backstage + Terraform + GitHub Actions. Onboarded 2,200+ engineers, reduced infra ticket volume by 78%, and cut new service time-to-prod from 3 weeks to 6 hours. Stack: Backstage, React, TypeScript, Terraform, AWS EKS, ArgoCD, Crossplane.

FinOps Automation Platform

2021 - 2022

Engineered intelligent cloud cost optimization system using AWS Cost Explorer APIs + ML anomaly detection. Automated rightsizing recommendations, spot instance orchestration, and waste detection saving \$7.2M in year 1. Open-sourced core components (1,400+ GitHub stars).

Akamai Edge Deploy

2018 - 2020

Designed global deployment pipeline for Akamai edge configuration updates. Leveraged canary deployments, automated rollback, and real-time traffic health monitoring across 350K+ edge servers. Reduced deployment-related incidents by 94%.

k8s-cost-analyzer (OSS)

2020 - Present

Open-source Kubernetes cost analysis tool with Prometheus integration. Provides per-namespace, per-pod cost attribution with optimization recommendations. 3,200+ GitHub stars, adopted by 180+ organizations including 3 Fortune 500 companies.

PUBLICATIONS & CONFERENCE TALKS

Book

"Infrastructure as Code: Patterns for Cloud-Native Enterprises" — O'Reilly

AWARDS AWS Community Hero (2022) CNCF Ambassador (2021–Present) Fidelity Innovation Award (2023) Akamai Engineering Excellence (2019) KubeCon Best Talk – Infra Track (2022)	Course "Mastering Terraform on AWS" — O'Reilly Learning Platform, 2021. 62,000+ enrolled learners.																																																																		
	Whitepaper "FinOps at Enterprise Scale" — AWS Whitepaper Series, 2022. Featured in AWS re:Invent keynote.																																																																		
	KubeCon NA "Zero-Downtime Multi-Region Kubernetes at Fidelity" — KubeCon NA 2022, Chicago. 1,200+ attendees. Video: 48K views.																																																																		
	AWS re:Invent "FinOps Automation: From Cost Chaos to Cost Intelligence" — AWS re:Invent 2023, Las Vegas.																																																																		
	HashiConf "Vault at Scale: Secrets Management for 4,000 Engineers" — HashiConf 2022, San Francisco.																																																																		
INTERESTS Open Source Contributing Competitive Rock Climbing Amateur Radio (KD1XYZ) 3D Printing & Hardware Mentoring / STEM Outreach	Blog Engineering blog: 45 articles, 280,000+ monthly page views. Topics: Kubernetes internals, FinOps, IaC patterns.																																																																		
	VOLUNTEERING & COMMUNITY																																																																		
	CNCF Ambassador Organizing Boston Cloud Native Meetup (2021–Present); 850+ members, monthly sessions. Coordinated 3 KubeCon scholarship applications for underrepresented engineers.																																																																		
	STEM Mentor Code.org and Hack.Diversity mentor; worked with 24 underrepresented students in Boston on cloud computing fundamentals. 18 students placed in tech internships.																																																																		
	Open Source Maintainer of 4 CNCF ecosystem tools (k8s-cost-analyzer, terraform-aws-modules/forks, helm-chart-releaser contrib). Reviewer for Kubernetes SIG-Autoscaling.																																																																		
PARSER EXTRACTION VERIFICATION TABLE																																																																			
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